nvfp4_gemv

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▼ Description

You will implement a batched matrix-vector multiplication kernel optimized for NVIDIA B200. To be explicit, you will be given a tuple of tensors:

```
(a, b, sfa, sfb, c)
```

where:

- `a` is M x K x L in K-major order in nvfp4(e2m1)
- `b` is 1 x K x L in K-major order in nvfp4(e2m1)
- `sfa` is M x (K // 16) x L in K-major order in fp8(e4m3fnuz)
- `sfb` is 1 x (K // 16) x L in K-major order in fp8(e4m3fnuz)
- `c` is M x 1 x L in fp16

Matrix sizes `M` is divisible by mma_tiler_mn[0] defined in the kernel, `K` is divisible by 64. The ranking criteria is the geometric mean of the benchmark results. For the grand price, your kernel will be evaluated against the speed of light analysis and the solution closest to the speed of light will be awarded the grand price.

```
The speed of light analysis based on the max(FFMA math throughput, DRAM mem
M    K     L time[us]
7168 16384 1 8.622
4096 7168  8 17.275
7168 2048  4 4.317
```

▼ Benchmark Shapes

- `{"k":16384,"l":1,"m":7168}`
- `{"k":7168,"l":8,"m":4096}`
- `{"k":2048,"l":4,"m":7168}`

RANKINGS REFERENCE SUBMISSION

Reference Implementation

```
import torch
from task import input_t, output_t
from utils import make_match_reference

# Scaling factor vector size
sf_vec_size = 16

# Helper function for ceiling division
def ceil_div(a, b):
    return (a + b - 1) // b


# Helper function to convert scale factor tensor to blocked format
def to_blocked(input_matrix):
    rows, cols = input_matrix.shape

    # Please ensure rows and cols are multiples of 128 and 4 respectively
    n_row_blocks = ceil_div(rows, 128)
    n_col_blocks = ceil_div(cols, 4)

    padded = input_matrix
    blocks = padded.view(n_row_blocks, 128, n_col_blocks, 4).permute(0, 2,
    rearranged = blocks.reshape(-1, 4, 32, 4).transpose(1, 2).reshape(-1, 3

    return rearranged.flatten()


def ref_kernel(
    data: input_t,
) -> output_t:
    """
    PyTorch reference implementation of NVFP4 block-scaled GEMV.
    """
    a_ref, b_ref, sfa_ref_cpu, sfb_ref_cpu, _, _, c_ref = data

    # Get dimensions from MxNxL layout
    _, _, l = c_ref.shape

    # Call torch._scaled_mm to compute the GEMV result
    for l_idx in range(l):
```

```
        # Convert the scale factor tensor to blocked format
        scale_a = to_blocked(sfa_ref_cpu[:, :, l_idx])
        scale_b = to_blocked(sfb_ref_cpu[:, :, l_idx])
        # (m, k) @ (n, k).T -> (m, n)
        res = torch._scaled_mm(
            a_ref[:, :, l_idx],
            b_ref[:, :, l_idx].transpose(0, 1),
            scale_a.cuda(),
            scale_b.cuda(),
            bias=None,
            out_dtype=torch.float16,
        )
        c_ref[:, 0, l_idx] = res[:, 0]
    return c_ref


def generate_input(
    m: int,
    k: int,
    l: int,
    seed: int,
):
    """
    Generate input tensors for NVFP4 block-scaled GEMV.

    Args:
        m: Number of rows in matrix A
        k: Number of columns in A (and length of vector b)
        l: Batch size
        seed: Random seed for reproducibility

    Returns:
        Tuple of (a, b, scale_a, scale_b, c) where:
            a: [m, k, l] - Input matrix in torch.float4e2m1fn_x2 data type
            b: [1, k, l] - Input vector in torch.float4e2m1fn_x2 data type
            scale_a: [m, k, l] - Input scale factors in torch.float8e4m3fn
            scale_b: [1, k, l] - Input scale factors in torch.float8e4m3fn
            scale_a_permuted: [32, 4, rest_m, 4, rest_k, l] - Input scale f
            scale_b_permuted: [32, 4, rest_n, 4, rest_k, l] - Input scale f
            c: [m, 1, l] - Output vector in torch.float16 data type
    """
    torch.manual_seed(seed)

    # GEMV N dimension is always 1
    n = 1
    # Scaling factor needs to pad the N size to 128
    n_padded_128 = 128
```

```
    # Generate uint8 tensor, then convert to float4e2m1fn_x2 data type
    a_ref = torch.randint(
        0, 4, (l, m, k // 2), dtype=torch.uint8, device="cuda"
    ).permute(1, 2, 0)
    # Pad b tensor's N dimension to 128 to call torch._scaled_mm for nvfp4
    b_ref = torch.randint(
        0, 4, (l, n_padded_128, k // 2), dtype=torch.uint8, device="cuda"
    ).permute(1, 2, 0)
    a_ref = a_ref.view(torch.float4_e2m1fn_x2)
    b_ref = b_ref.view(torch.float4_e2m1fn_x2)

    # Create float16 output tensor
    c_ref = torch.randn((l, m, n), dtype=torch.float16, device="cuda").perm
        1, 2, 0
    )

    # Helper function to prepare the scale factor tensors for both referenc
    # kernel and customize kernel. The customized data layout can be found
    # https://docs.nvidia.com/cuda/cublas/index.html?highlight=fp4#d-block-
    def create_scale_factor_tensors(l, mn, sf_k):
        # Create the reference scale factor tensor (mn, sf_k, l) on CPU.
        ref_shape = (l, mn, sf_k)
        ref_permute_order = (1, 2, 0)
        # Init with uint8 tensor, then convert to float8_e4m3fn
        ref_f8_random_int = torch.randint(0, 3, ref_shape, dtype=torch.int8
        ref_f8_torch_tensor = ref_f8_random_int.to(dtype=torch.float8_e4m3f
        # permute to match ref_permute_order
        ref_f8_torch_tensor_permuted = ref_f8_torch_tensor.permute(*ref_per

        atom_m = (32, 4)
        atom_k = 4
        mma_shape = (
            l,  # batch size
            ceil_div(mn, atom_m[0] * atom_m[1]),
            ceil_div(sf_k, atom_k),
            atom_m[0],
            atom_m[1],
            atom_k,
        )

        # Reorder scale factor tensor to (32, 4, rest_m, 4, rest_k, l) layo
        # Which is needed by the CuTe customized kernel
        mma_permute_order = (3, 4, 1, 5, 2, 0)
        # Generate a random int8 tensor, then convert to float8_e4m3fn
        rand_int_tensor = torch.randint(0, 3, mma_shape, dtype=torch.int8,
        reordered_f8_torch_tensor = rand_int_tensor.to(dtype=torch.float8_e
        # Permute according to mma_permute_order
        reordered_f8_torch_tensor = reordered_f8_torch_tensor.permute(*mma_
```

```
        # GPU-side vectorized reordering (replaces slow CPU nested loops)
        # Create index grids for all dimensions
        i_idx = torch.arange(mn, device='cuda')
        j_idx = torch.arange(sf_k, device='cuda')
        b_idx = torch.arange(l, device='cuda')

        # Create meshgrid for all combinations of (i, j, b)
        i_grid, j_grid, b_grid = torch.meshgrid(i_idx, j_idx, b_idx, indexi

        # Calculate target indices in vectorized manner
        mm = i_grid // (atom_m[0] * atom_m[1])
        mm32 = i_grid % atom_m[0]
        mm4 = (i_grid % 128) // atom_m[0]
        kk = j_grid // atom_k
        kk4 = j_grid % atom_k

        # Perform the reordering with advanced indexing (all on GPU)
        reordered_f8_torch_tensor[mm32, mm4, mm, kk4, kk, b_grid] = ref_f8_

        return ref_f8_torch_tensor_permuted.cpu(), reordered_f8_torch_tenso

    sf_k = ceil_div(k, sf_vec_size)
    sfa_ref_cpu, sfa_permuted = create_scale_factor_tensors(l, m, sf_k)
    sfb_ref_cpu, sfb_permuted = create_scale_factor_tensors(l, n_padded_128

    sfa_ref = sfa_ref_cpu.to("cuda")
    sfb_ref = sfb_ref_cpu.to("cuda")

    return (a_ref, b_ref, sfa_ref, sfb_ref, sfa_permuted, sfb_permuted, c_r


check_implementation = make_match_reference(ref_kernel, rtol=1e-03, atol=1e
```

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